Course Title: Database Systems

Course Code: CSE302

Section: 01

Semester: Spring 2025

Relational Modelling and Normalization Techniques

Prepared by:

Name	ID no.
Raquib Uddin Sarkar	2023-2-60-101
Md. Arifur Rahman Shihab	2023-2-60-176
Md Fahab bin Abu bakar	2023-3-60-075
Juboraj Podder	2023-2-60-048

Prepared For:

Dr. Mohammad Rezwanul Huq

Associate Professor

Department of Computer Science and Engineering

East West University

Date of submission: 1 June, 2025

Table of Contents

Title	Page no.
Normalization	3
1. Venue	3
2. Match	4
3. Match_Result_Details	5
4. Match_Team_Detail	6
5. Match_Umpire_Details	7
6. Team	8
7. Player	9
8. Player_Role	10
9. Umpire	11
10. Innings	12
11. Delivery	13-14
12. Wickets	14-15

Normalization

1. Venue

Column Name	Data Type	Nullable	Default	Primary Key...
VENUE_NAME	VARCHAR2(50 BYTE)	N		1
CITY	VARCHAR2(50 BYTE)	Y		
CAPACITY	NUMBER	Y		
VENUEPHOTO	BLOB	Y		
FILE_NAME	VARCHAR2(10 BYTE)	Y		
MIME_TYPE	VARCHAR2(10 BYTE)	Y		
UPDATE_DATE	VARCHAR2(20 BYTE)	Y		
CHARACTER_SET	VARCHAR2(10 BYTE)	Y		

Venue table is in BCNF.

Primary Key: Venue_name

Non-Key columns: city, capacity, venuephoto, file_name, mime_type, update_date, character_set

Functional Dependencies:

Venue_name → city, capacity, venuephoto, file_name, mime_type, update_date, character_set

(All non-key attributes depend on venue_name, which is the primary key)

BCNF Check:

- **1NF (First Normal Form):** There is no multivalued attributes. That is why Venue is 1NF satisfied.
- **No Partial Dependencies:** Since there's a single-column primary key (venue_name), there are no partial dependencies (already satisfies 2NF).
- **No Transitive Dependencies:** None of the non-key attributes (e.g., city, capacity, etc.) depend on other non-key attributes, satisfying 3NF.
- **BCNF Rule**: The only functional dependency is venue_name determining all other attributes, and so here venue_name is a candidate key.

2. **Match**

Column Name	Data Type	Nullable	Default	Primary Key...
MATCH_NO	NUMBER	N		1
Date	DATE	Y		
MATCH_TYPE	VARCHAR2(20 BYTE)	N		
VENUE_NAME	VARCHAR2(50 BYTE)	N		

Match table is in BCNF.

Primary Key: Match_No

Foreign Keys:

- Venue_name (FK, referencing venue table)

Non-Key columns: date, match_type

Functional Dependencies:

Match_No → date, match_type,venue_name

Foreign Key Dependencies:

- Venue_venue_name → **Venue_name**

BCNF Check:

- **1NF (First Normal Form):** No multivalued attributes, so 1NF satisfied.
- **No Partial Dependencies:** Since there's a single-column primary key (match_no), there are no partial dependencies (already satisfies 2NF).
- **No Transitive Dependencies:** None of the non-key attributes (e.g. date,match_type) depend on other non-key attributes, satisfying 3NF.
- **BCNF Rule**: Here match_no is the only functional dependency which is determining all other attributes. That is why match_no is a candidate key.

3. **Match_Result_Details**

Column Name	Data Type	Nullable	Default	Primary Key
MATCH_NO	NUMBER	N		1
TOSS_WINNER	VARCHAR2(30 BYTE)	N		
TARGET	NUMBER	Y		
MATCH_WINNER	VARCHAR2(30 BYTE)	N		
Win_By(Runs/Wickets)	VARCHAR2(10 BYTE)	Y		
RESULT_MARGIN	NUMBER	Y		
PLAYER_OF_THE_MATCH	VARCHAR2(50 BYTE)	N		
METHOD	VARCHAR2(5 BYTE)	Y		

Match_Result_Details table is in BCNF.

Primary Key: match_no

Foreign Keys:

- toss_winner(references to team)
- match_winner(references to team)
- player_of_the_match (references to player)
-

Non-Key columns:
toss_winner,target,match_winner,win_by,result_margin,player_of_the_match,method

Functional Dependencies:

match_no → toss_winner,target,match_winner,win_by,result_margin,player_of_the_match,method

Foreign Key Dependencies:

- team_name → toss_winner
- team_name → match_winner
- player_name → player_of_the_match

It is a weak entity set. Because it doesn't have it's own primary key. It borrows the primary key of the identifying entity set which is match.

BCNF Check:

- **1NF (First Normal Form):** No multivalued attributes, so 1NF satisfied.
- **No Partial Dependencies:** Since there's a single-column primary key (match_no), there are no partial dependencies (already satisfies 2NF).
- **No Transitive Dependencies:** None of the non-key attributes (target, method etc) depend on other non-key attributes, satisfying 3NF.

BCNF Rule: Here match_no is the only functional dependency which is determining all other attributes. That is why match_no is a candidate key.

4. **Match_Team_Detail**

Column Name	Data Type	Nullable	Default	Primary Key
MATCH_NO	NUMBER	N		1
TEAM_NAME_1	VARCHAR2(50 BYTE)	N		
TEAM_NAME_2	VARCHAR2(50 BYTE)	N		
TOSS_DECISION	VARCHAR2(10 BYTE)	Y		

Match_Team_Details table is in BCNF.

Primary Key: match_no

Foreign Keys:

- Team_Name_1(references to team)
- Team_Name_2(references to team)

Non-Key columns: toss_decision

Functional Dependencies:

match_no → Team_Name_1, Team_Name_2, toss_decision

Foreign Key Dependencies:

- team_name → Team_Name_1
- team_name → Team_Name_2

It is a weak entity set. Because it doesn't have its own primary key. It borrows the primary key of the identifying entity set which is match.

BCNF Check:

- **1NF (First Normal Form):** No multivalued attributes, so 1NF satisfied.
- **No Partial Dependencies:** Since there's a single-column primary key (match_no), there are no partial dependencies (already satisfies 2NF).
- **No Transitive Dependencies:** None of the non-key attributes (toss_decision) depend on other non-key attributes, satisfying 3NF.

BCNF Rule: Here match_no is the only functional dependency which is determining all other attributes. That is why match_no is a candidate key.

5. **Match_Umpire_Details**

Column Name	Data Type	Nullable	Default	Primary Key
MATCH_NO	NUMBER	N		1
UMPIRE_NAME_1	VARCHAR2(50 BYTE)	N		
UMPIRE_NAME_2	VARCHAR2(50 BYTE)	N		

Match_Umpire_Details table is in BCNF.

Primary Key: match_no

Foreign Keys:

- Umpire_Name_1(references to Umpire)
- Umpire_Name_2(references to Umpire)

Non-Key columns:

Functional Dependencies:

match_no → Umpire_Name_1, Umpire_Name_2

Foreign Key Dependencies:

- Umpire_name → Umpire_Name_1
- Umpire_name → Umpire_Name_2

It is a weak entity set. Because it doesn't have its own primary key. It borrows the primary key of the identifying entity set which is match.

BCNF Check:

- **1NF (First Normal Form):** No multivalued attributes, so 1NF satisfied.
- **No Partial Dependencies:** Since there's a single-column primary key (match_no), there are no partial dependencies (already satisfies 2NF).
- **No Transitive Dependencies:** None of the non-key attributes depend on other non-key attributes, satisfying 3NF.

BCNF Rule: Here match_no is the only functional dependency which is determining all other attributes. That is why match_no is a candidate key.

6. Team

Column Name	Data Type	Nullable	Default	Primary Key
TEAM_NAME	VARCHAR2(30 BYTE)	N		1
TEAM_INITIAL	VARCHAR2(10 BYTE)	Y		
VENUE_VENUE_NAME	VARCHAR2(50 BYTE)	N		
TEAMICON	BLOB	Y		
FILE_NAME	VARCHAR2(10 BYTE)	Y		
MIME_TYPE	VARCHAR2(10 BYTE)	Y		
UPDATE_DATE	VARCHAR2(20 BYTE)	Y		
CHARACTER_SET	VARCHAR2(10 BYTE)	Y		

Team table is in BCNF.

Primary Key: Team-Name

Foreign Keys:

- Venue_Name_1(references to Venue)

Functional Dependencies: Team_Name → team_initial, venue_venue_name(home_ground),teamIcon,file_name,mime_type,update_date, character_set

Foreign Key Dependencies:

Venue_Name →Venue_venue_name(Home_Ground)

BCNF Check:

- **1NF (First Normal Form):** All columns contain atomic (indivisible) values, with no repeating groups or arrays, and each record is unique, identified by the primary key team_name.
- **No Partial Dependencies:** The table has a single-column primary key (team_name), and all non-key attributes (team_initial, home_ground etc) depend fully on the primary key (no violation of 2NF).
- **No Transitive Dependencies:** None of the non-key attributes (team_initial, home_ground etc) depend on other non-key attributes., so the table satisfies 3NF.
- **BCNF Rule**: The only functional dependency is team_name determining all other attributes, and team_name is a candidate key.

7. **Player**

Column Name	Data Type	Nullable	Default	Primary Key
PLAYER_NAME	VARCHAR2(50 BYTE)	N		1
DATE_OF_BIRTH	DATE	Y		
NATIONALITY	VARCHAR2(50 BYTE)	Y		
TEAM_NAME	VARCHAR2(50 BYTE)	N		
PLAYERPHOTO	BLOB	Y		
FILE_NAME	VARCHAR2(10 BYTE)	Y		
MIME_TYPE	VARCHAR2(10 BYTE)	Y		
UPDATE_DATE	VARCHAR2(20 BYTE)	Y		
CHARACTER_SET	VARCHAR2(10 BYTE)	Y		

Player table is in BCNF.

Primary Key: Player_Name

Foreign Keys:

- Team_Name(references to team)

Functional Dependencies: Player_Name → date_of_birth, nationality, Team-name, playerPhoto, file_name,mime_type,update_date, character_set

Foreign Key Dependencies:

Team_Name →Team_name

BCNF Check:

- **1NF (First Normal Form):** All columns contain atomic (indivisible) values, with no repeating groups or arrays, and each record is unique, identified by the primary key player_name.
- **No Partial Dependencies:** The table has a single-column primary key (player_name), and all non-key attributes (nationality, date_of_birth etc) depend fully on the primary key (no violation of 2NF).
- **No Transitive Dependencies:** None of the non-key attributes (nationality, date_of_birth etc) depend on other non-key attributes., so the table satisfies 3NF.

BCNF Rule: The only functional dependency is player_name determining all other attributes, and player_name is a candidate key.

8. Player_Role

Column Name	Data Type	Nullable	Default	Primary Key
PLAYER_NAME	VARCHAR2(50 BYTE)	N		1
ROLE	VARCHAR2(50 BYTE)	Y		
Batting_Style(Left/Right)	VARCHAR2(20 BYTE)	Y		
Bowling_Style(Left/Right)	VARCHAR2(50 BYTE)	Y		
BATTING_TYPE	VARCHAR2(50 BYTE)	Y		

Player_Role table is in BCNF.

Primary Key: player_name

Non-Key columns: role, batting_style, bowling_style, batting_type

Functional Dependencies:

player_name → role, batting_style, bowling_style, batting_type

It is a weak entity set. Because it doesn't have its own primary key. It borrows the primary key of the identifying entity set which is player.

BCNF Check:

- **1NF (First Normal Form):** No multivalued attributes, so 1NF satisfied.
- **No Partial Dependencies:** Since there's a single-column primary key (player_name), there are no partial dependencies (already satisfies 2NF).
- **No Transitive Dependencies:** None of the non-key attributes depend on other non-key attributes, satisfying 3NF.

BCNF Rule: Here player_name is the only functional dependency which is determining all other attributes. That is why player_name is a candidate key.

9. **Umpire**

Column Name	Data Type	Nullable	Default	Primary Key
UMPIRE_NAME	VARCHAR2(50 BYTE)	N		1
NATIONALITY	VARCHAR2(50 BYTE)	Y		
AGE	NUMBER	Y		

Umpire table is in BCNF.

Primary Key: Umpire_Name

Functional Dependencies: **Umpire_Name** → nationality, age

BCNF Check:

- **1NF (First Normal Form):** All columns contain atomic (indivisible) values, with no repeating groups or arrays, and each record is unique, identified by the primary key umpire_name.
- **No Partial Dependencies:** The table has a single-column primary key (umpire_name), and all non-key attributes (nationality, age) depend fully on the primary key (no violation of 2NF).
- **No Transitive Dependencies:** None of the non-key attributes (nationality, age) depend on other non-key attributes., so the table satisfies 3NF.

BCNF Rule: The only functional dependency is umpire_name determining all other attributes, and umpire_name is a candidate key.

10. Innings

+ Add Column | Modify Column | Rename Column | Drop Column | UI Defaults | Refresh | More

Column Name	Data Type	Nullable	Default	Primary Key
INNING_NO	NUMBER	N		1
MATCH_NO	NUMBER	N		2
WICKETS	NUMBER	Y		
TOTAL_RUNS	NUMBER	Y		
OVERS	NUMBER	Y		

Innings table is in 3NF.

Primary Key: Inning_no, Match_No

Non-Key columns: wickets, total_runs, overs

Foreign Keys:

- Match_No(references to match)

Functional Dependencies:

Inning_No, Match_No→ wickets, total_runs, overs

Foreign Key Dependencies:

Match_No → Match_No

3NF Check:

- **1NF (First Normal Form):** No multivalued attributes, so 1NF satisfied.
- **No Partial Dependencies:** Though there's a composite primary key (Inning_No, Match_No), there are no partial dependencies (already satisfies 2NF).
- **No Transitive Dependencies:** None of the non-key attributes depend on other non-key attributes, satisfying 3NF.

11. Delivery

Column Name	Data Type	Nullable	Default	Primary Key
DELIVERY_ID	NUMBER	N		1
MATCH_NO	NUMBER	Y		
INNING_NO	NUMBER	Y		
OVER	NUMBER	N		
BALL	NUMBER	N		
BOWLER	VARCHAR2(50 BYTE)	N		
BATTER	VARCHAR2(50 BYTE)	N		
NON_STRIKER	VARCHAR2(50 BYTE)	N		
BATSMAN_RUNS	NUMBER	Y		
FIELDER	VARCHAR2(50 BYTE)	Y		
EXTRA_TYPE	VARCHAR2(20 BYTE)	Y		
EXTRA_RUNS	NUMBER	Y		

Delivery table is in BCNF.

Primary Key: Delivery_ID

Non-Key columns: over, ball,batsman_runs,extra_type,Extra_runs

Foreign Keys:

- Match_No(references to match)
- Inning_no (references to innings)
- Bowler(references to player)
- Batter(references to player)
- Non_Striker(references to player)
- Fielder(references to player)

Functional Dependencies:

Delivery_ID → over, ball,batsman_runs,extra_type,Extra_runs, Match_No, Inning_no, Bowler, Batter, Non_Striker, Fielder

Foreign Key Dependencies:

- Match_No → Match_No
- Inning_no → Inning_no
- Bowler → player_name
- Batter → player_name
- Non_Striker → player_name
- Fielder → player_name

BCNF Check:

- **1NF (First Normal Form):** All columns contain atomic (indivisible) values, with no repeating groups or arrays, and each record is unique, identified by the primary key delivery_id.
- **No Partial Dependencies:** The table has a single-column primary key (delivery_id), and all non-key attributes (over, ball etc) depend fully on the primary key (no violation of 2NF).
- **No Transitive Dependencies:** None of the non-key attributes (over, ball etc) depend on other non-key attributes., so the table satisfies 3NF.
- **BCNF Rule**: The only functional dependency is delivery_id determining all other attributes, and delivery_id is a candidate key.

12. Wickets

Column Name	Data Type	Nullable	Default	Primary Key
MATCH_MATCH_NO	NUMBER	N		
DELIVERY_DELIVERY_ID	NUMBER	N		
WICKET_ID	NUMBER	N		1
DISMISSAL_KIND	VARCHAR2(50 BYTE)	Y		
PLAYER_PLAYER_NAME	VARCHAR2(50 BYTE)	Y		

Wickets table is in BCNF.

Primary Key: Wicket_ID

Non-Key columns: dismissal_kind

Foreign Keys:

- Match_Match_No(references to match)
- Delivery_Delivery_ID (references to delivery)
- Player_Player_Name(references to player)

Functional Dependencies:

Wicket_ID → Match_Match_No, Delivery_Delivery_ID, Player_Player_Name(Dismissed_Player), dismissal_kind

Foreign Key Dependencies:

- Match_Match_No → Match_No
- Delivery_Delivery_ID → Delivery_ID
- Player_Player_Name → player_name(dismissed_player)

BCNF Check:

13. **1NF (First Normal Form):** All columns contain atomic (indivisible) values, with no repeating groups or arrays, and each record is unique, identified by the primary key Wicket_ID.
14. **No Partial Dependencies:** The table has a single-column primary key (Wicket_ID), and all non-key attributes (dismissal_kind) depend fully on the primary key (no violation of 2NF).
15. **No Transitive Dependencies:** None of the non-key attributes (dismissal_kind) depend on other non-key attributes., so the table satisfies 3NF.
16. **BCNF Rule**: The only functional dependency is Wicket_ID determining all other attributes, and Wicket_ID is a candidate key.